

Cartesian space control in SMC

Marko Guberina

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1 Introduction

There is nothing new or special about Cartesian control in SMC. This text is completely written in reference to

- blog posts by [Stéphane Caron](#), in particular:
 - [differential inverse kinematics](#)
 - [Jacobian of kinematic task](#)
- Frank C. Park, Kevin Lynch - Modern Robotics
- Murray, R.M., Li, Z. and Sastry, S.S., 2017. A mathematical introduction to robotic manipulation. CRC press

However, it is useful to have a summary at arms length.

1.1 Notation used

- $q \in \mathbb{R}^{nq}$ - vector of joint positions
- $v \in \mathbb{R}^{nv}$ - vector of joint velocities
- $a \in \mathbb{R}^{nv}$ - vector of joint acceleration
- $T_a^b \in SE(3)$ - transformation from frame $\{a\}$ to $\{b\}$, equivalently frame $\{b\}$ as seen in frame $\{a\}$
- $V \in se(3)$ - twist, i.e. 6D velocity, $V = \begin{bmatrix} v & \omega \end{bmatrix}^T$

2 Differential inverse kinematics

Skipping all prerequisites on geometry and robot kinematics, let's start with

$$J(q)v = V \tag{1}$$

where $J \in \mathbb{R}^{6 \times nv}$ is the Jacobian (linearization) of the forward kinematics map, i.e. $J(q) = \frac{\partial FK}{\partial q}$ and V is end-effector twist.

Usually J is not invertible, and we do not assume it to be invertible in this text. Hence the task of inverse kinematics is to solve

$$\min_v \|Jv - V\|^2 \quad (2)$$

We can also throw some scaling in with $K_p V$, with $K_p \in [0, 1]$. If this was all there was to it, we would solve this the pseudoinverse and call it a day, i.e.

$$v = J^T(JJ^T + \lambda I)V = J^\dagger V \quad (3)$$

But if we have, for example, linear constraints, we can interpret the equation as a quadratic program (QP) solve it with a QP solver.

Moreover, we very often have additional tasks, or want to use the redundant joints of freedom for more graceful movement (exploit the nullspace of J). This is especially important if there's multiple frames we care about, for example in a mobile manipulator. The additional tasks can be simply summed together:

$$\min_v \sum_i w_i \|J_i v - K_i V_i\| \quad (4)$$

Solving this will result in a compromise between the two targets, the amount of compromising being determined with the weights w_i . Famously, this emulates the behavior of a hierarchical quadratic program (HQP).

2.1 Inequality constraints

The most important inequality constraint concerns joint limits, making the problem look like

$$\min_v \sum_i w_i \|J_i v - K_i V_i\| \quad (5)$$

$$\text{s.t. } v_{\min} \leq v \leq v_{\max} \quad (6)$$

The same can be done with joint angle limits $q_{\min} \leq q \leq q_{\max}$.

2.2 Transcribing into standard QP form

2.3 Common secondary tasks

2.3.1 Collision avoidance via control-barrier functions

2.3.2 Manipulability maximization